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Live Virus EV-A71 - RD - Antiviral Screening Assay

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Protocol status: Working

We use this protocol and it's working

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Disclaimer

Assay is performed under biosafety level 3 conditions.

Abstract

The following is a protocol for live virus antiviral screening against EV-A71 in vitro. Cells are prophylactically treated with antiviral candidates that have been serially diluted to investigate activity, via IC50 values generated via analysis of immunofluorescent staining, against EV-A71.

Each live virus screening assay is accompanied by a matching cytotoxicity screening assay in uninfected cells to investigate drug toxicity.

Materials

10% Media

2% Media

96-well plates

96-well deep well plates

12-channel 200/300ul multichannel pipette

200/300ul filtered tips

Compounds of interest

Virus(es) of interest

PPE

reservoir

Protocol materials

 Deep Well Plate (96 well) **Thermo Fisher Catalog #10045**

Safety warnings

 Please be sure to wear proper Personal Protective Equipment (PPE) while performing this experiment.

Before start

Always wear appropriate PPE for this protocol

Refer to Material Safety Data Sheets for additional safety and handling information.

Seeding

- 1 The day before your assay, seed 96-well plates with RD cells at **5,000 cells/well** using standard RD growth media



You will need to make **one pair of plates** for every 3 compounds. One plate is the antiviral plate that will be infected, and the other is the cytotoxicity plate that will measure the toxicity of the antiviral candidates.

For each pair you can test 3 compounds (in triplicate, 8 dilutions per compound), 2 DMSO controls, and 1 uninfected control in the following layout:

DMSO	Comp. 1 Max	Comp. 1 Max	Comp. 1 Max	Comp. 2 Max	Comp. 2 Max	Comp. 2 Max	Comp. 3 Max	Comp. 3 Max	Comp. 3 Max	DMSO	Uninfected Control
DMSO	1:3	1:3	1:3	1:3	1:3	1:3	1:3	1:3	1:3	DMSO	Uninfected Control
DMSO	1:3	1:3	1:3	1:3	1:3	1:3	1:3	1:3	1:3	DMSO	Uninfected Control
DMSO	1:3	1:3	1:3	1:3	1:3	1:3	1:3	1:3	1:3	DMSO	Uninfected Control
DMSO	1:3	1:3	1:3	1:3	1:3	1:3	1:3	1:3	1:3	DMSO	Uninfected Control
DMSO	1:3	1:3	1:3	1:3	1:3	1:3	1:3	1:3	1:3	DMSO	Uninfected Control
DMSO	1:3	1:3	1:3	1:3	1:3	1:3	1:3	1:3	1:3	DMSO	Uninfected Control
DMSO	1:3	1:3	1:3	1:3	1:3	1:3	1:3	1:3	1:3	DMSO	Uninfected Control

Treatment layout

Protocol

NAME

Cytotoxicity Screening Assay - Paired with Antiviral Assays

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Setting up the conditions

- 2 Create a screening priority list ahead of time. This list will have the compounds of interest, what maximum concentration you'd like to screen at, and where the compounds are located. We recommend organizing the compounds the day before.
- 3 The serial dilutions will follow different protocols depending on if you will do them by hand or by using a



Equipment

Tecan D300e

Liquid Handler

Tecan

30100152

https://lifesciences.tecan.com/products/liquid_handling_and_automation/tecan_d300e_

Before the assay, you should prepare enough 2% media **supplemented with DMSO** for your screening. **DMSO supplementation will correspond to the highest amount of drug stock used.**

STEP CASE

Manual 12 steps

Performing the serial dilutions manually.

If your lab does not have a Tecan D300e machine you will need to do the dilutions by hand.

- 4 Remove your compounds from -20 to thaw while you set up.
- 5 Prepare the  Deep Well Plate (96 well) Thermo Fisher Catalog #10045 according to this diagram:



Deep Well Plate X (without **tecan**)

Comp. 1 Max	Comp. 2 Max	Comp. 3 Max	Comp. 4 Max	Comp. 5 Max	Comp. 6 Max	Comp. 7 Max	Comp. 8 Max	Comp. 9 Max	Comp. 10 Max	Comp. 11 Max	Comp. 12 Max
(1100ul)	(1110ul)	(1110ul)	(1110ul)	(1110ul)	(1110ul)	(1110ul)	(1110ul)	(1110ul)	(1110ul)	(1110ul)	(1110ul)
(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)
(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)
(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)
(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)
(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)
(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)
(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)	(750ul)

Deep well set up for manual serial dilutions

The first row will be the [max] of your compounds. This row will contain **1100ul** of 2% media **without DMSO**. The media for this row does not receive any DMSO since you will add drug dissolved in DMSO directly to it.

The remaining rows of the plate will be filled with **750ul 2% media + supplemented with DMSO**.

These volumes are enough to treat both the antiviral and cytotoxicity plates.

- 6 Perform the serial dilutions:
Add the compounds of interest to the first row of the deep well (containing 1100ul media).

- 6.1 **mix well** and move **375ul** from row 1 to the row 2. This is your first 1:3 dilution.



Deep Well Plate X (without **tecan**)

Comp. 1 Max	Comp. 2 Max	Comp. 3 Max	Comp. 4 Max	Comp. 5 Max	Comp. 6 Max	Comp. 7 Max	Comp. 8 Max	Comp. 9 Max	Comp. 10 Max	Comp. 11 Max	Comp. 12 Max
(1100ul)	(1110ul)	(1110ul)	(1110ul)	(1110ul)	(1110ul)	(1110ul)	(1110ul)	(1110ul)	(1110ul)	(1110ul)	(1110ul)
(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)
(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)
(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)
(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)
(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)
(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)
(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)	(750ul+375ul)

Schematic of the manual serial dilutions

- 6.2 Repeat until the entire deep well has been done.
You now have 8 concentrations of your compounds of interest.



- 7 Remove the growth media from the plates. Work one pair of plates at a time to prevent drying of the cells.

Add **100ul** of the serial dilutions to the wells in the following set up: You will perform this for **both** the infection plate and the matching cytotoxicity plate

DMSO	Comp. 1 Max	Comp. 1 Max	Comp. 1 Max	Comp. 2 Max	Comp. 2 Max	Comp. 2 Max	Comp. 3 Max	Comp. 3 Max	Comp. 3 Max	DMSO	Uninfected Control
DMSO	1:3	1:3	1:3	1:3	1:3	1:3	1:3	1:3	1:3	DMSO	Uninfected Control
DMSO	1:3	1:3	1:3	1:3	1:3	1:3	1:3	1:3	1:3	DMSO	Uninfected Control
DMSO	1:3	1:3	1:3	1:3	1:3	1:3	1:3	1:3	1:3	DMSO	Uninfected Control
DMSO	1:3	1:3	1:3	1:3	1:3	1:3	1:3	1:3	1:3	DMSO	Uninfected Control
DMSO	1:3	1:3	1:3	1:3	1:3	1:3	1:3	1:3	1:3	DMSO	Uninfected Control
DMSO	1:3	1:3	1:3	1:3	1:3	1:3	1:3	1:3	1:3	DMSO	Uninfected Control
DMSO	1:3	1:3	1:3	1:3	1:3	1:3	1:3	1:3	1:3	DMSO	Uninfected Control
DMSO	1:3	1:3	1:3	1:3	1:3	1:3	1:3	1:3	1:3	DMSO	Uninfected Control

Treatment layout of the HeLa-ACE2 plates.

Notice how the inner columns of the plate receive experimental treatment, while the outer columns are the controls.

You should always have the plates in the correct orientation (top left corner should be A1) so that you are sure where the highest and lowest [compound] are.

- 8 Incubate the cells at 37 °C 5% CO2 for 02:00:00 .

2h



This pretreatment time frame could differ depending on what sort of compounds you'd like to test.

Prepping for infection

1h 30m

- 9 Dilute EV-A71 in 2% media to **MOI 0.5**





Protocol

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Calculating Multiplicity of Infection (MOI)

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- 10 perform the **mock infection on the cytotoxicity plates**. We want both plates to match so that we can compare them later.

Add **50ul 2% media** to the wells.

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Cytotoxicity Screening Assay - Paired with Antiviral Assays

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Infection

1d

- 11 Infect the wells with 50ul

- 12 Incubate at  37 °C for **7 days**

1d



MTS



- 13 Instead of fixing with formaldehyde, use the MTS protocol at the end of the cytotoxicity assay for **both** the antiviral and cytotoxicity plates

Protocol

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Cytotoxicity Screening Assay - Paired with Antiviral Assays

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